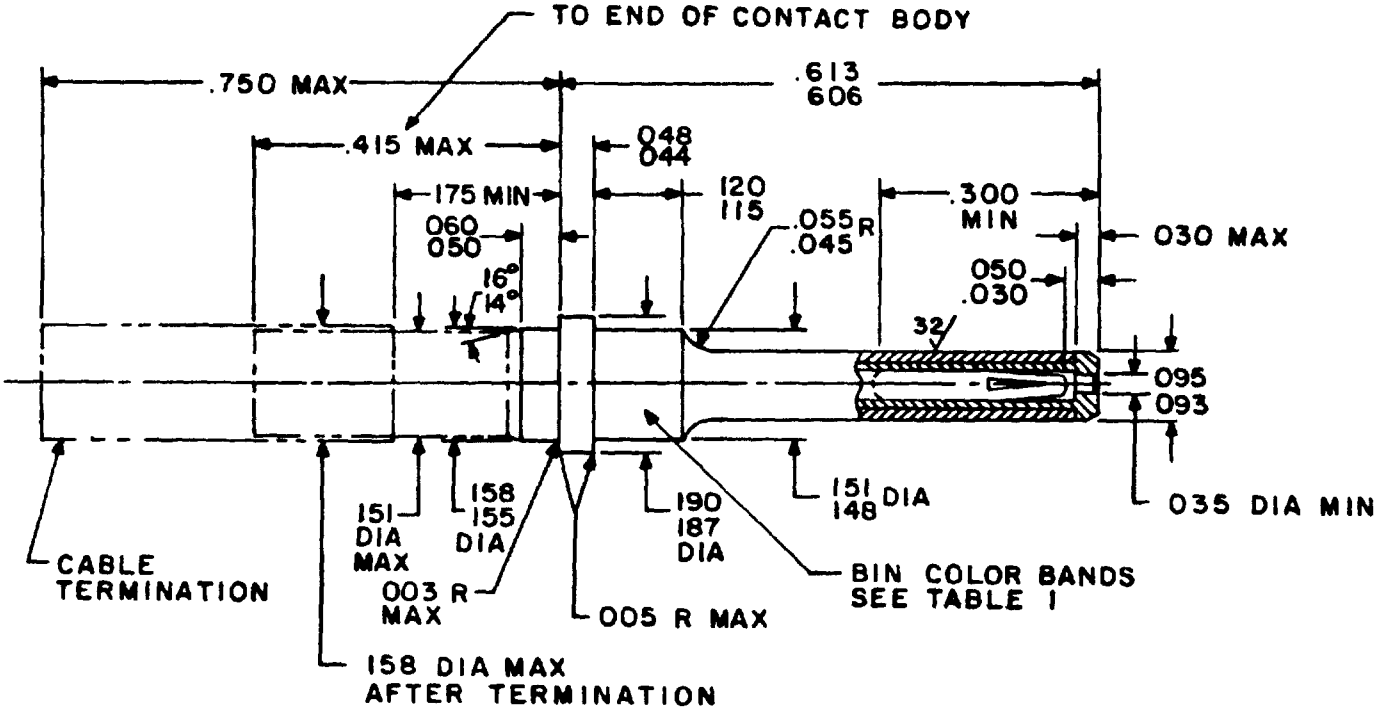
MILITARY SPECIFICATION SHEET

CONTACTS, ELECTRICAL CONNECTORS, PIN, SOLDER, © REMOVABLE, SHIELDED, SIZE 12 (FOR MIL-C-83723 SERIES III, MIL-C-26482 SERIES 2, MIL-C-83733 AND DOD-C-83527 CONNECTORS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the contacts described herein shall consist of this specification and the latest issue of MIL-C-39029



Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm
003	0.08	.045	1 14	.093	2.36	.151	3 84	.190	4 83
.005	0.13	.048	1.22	.095	2 41	.155	3.94	.300	7.62
.030	0.76	.050	1 27	.115	2 92	.158	4.01	.415	10.54
.035	0.89	.055	1.40	120	3.05	.175	4 44	606	15.39
.044	1.12	.060	1.52	.148	3.76	187	4.75	613	15.57
								750	19.05

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimensions shown apply after plating.
4. When this contact is used in MIL-C-83723 series III connector, intermateability can not be achieved with MIL-C-26500 connectors.

FIGURE 1. Connector contact.

© denotes changes

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:41Z
Check the source to verify that this is the current version before use.

ASSEMBLY INSTRUCTIONS:

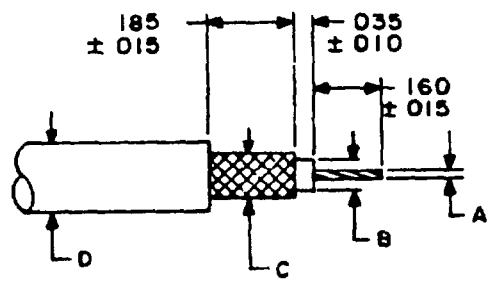
(C) 1. Strip cable as shown on figure 2.

2. Insert cable into contact until it is fully bottomed.

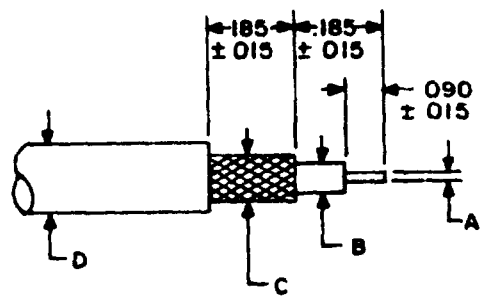
3. Heat contact with appropriate tools as shown on table II until solder melts and flows and strain relief tubing conforms to cable. Using designated heating tools will facilitate proper soldering without endangering cable insulation integrity.

(C) 4. If sealing via rear grommet is required for twisted pairs (BIN 401), an additional sealing device is required.

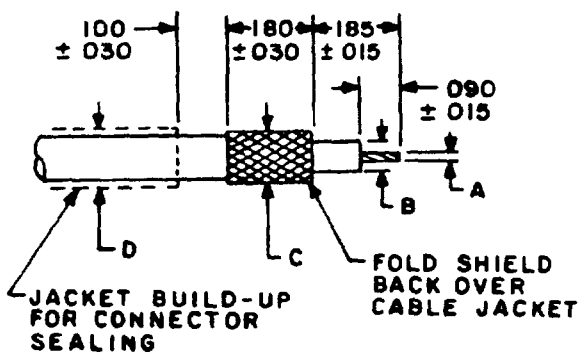
(C) 5. See tables I, IV, and V for cable accommodation.



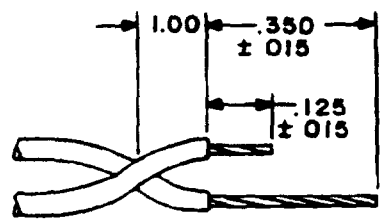
(C) CABLE PREPARATION FOR BIN 399



(C) CABLE PREPARATION FOR BIN 400
STRAIGHT SHIELD



(C) CABLE PREPARATION FOR BIN 400
FOLDBACK SHIELD



(C) CABLE PREPARATION FOR BIN 401

Inches	mm
.010	0.25
.015	0.38
030	0.76
.035	0.89
.090	2.29
.100	2.54
125	3.18
.160	4.06
.180	4.57
.185	4.70
.350	8 89
1 00	25 4

FIGURE 2. Assembly instructions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:41Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Design and construction:
Dimensions and configuration: See figure 1 and table I.

(C) TABLE I. Design and characteristics.

BIN code	Color band			Cable accommodation (see figure 2)				Shield preparation technique	Test cables	Type	Class
	1ST	2ND	3RD	1/ A	2/ B	3/ C	4/ D				
399	Orange	White	White	.026 Max. .011 Min.	.105 Max. .093 Min.	.124 Max. .112 Min.	.145 Max. .120 Min.	Straight	M17/095-RG180	D	A
400	Yellow	Black	Black	.026 Max. .011 Min.	.081 Max. .055 Min.	.099 Max. .077 Min.	.140 Max. .094 Min.	Straight	M17/094-RG179 Raychem 9530H1024-9 5/		
				.026 Max. .011 Min.	.045 Max. .030 Min.	.099 Max. .077 Min. after fold-back	.145 Max. .094 Min. after build-up	Foldback	M17/093-RG178		
401	Yellow	Black	Brown	24 and 26 AWG twisted pair				N/A	Twisted pair 24 and 26 AWG per MIL-C-55021		

1/ A: Conductor diameter.
2/ B: Dielectric diameter.
3/ C: Shield diameter.
4/ D: Jacket diameter.
5/ Or equivalent.

Inches	mm	Inches	mm
.011	0.28	.099	2.51
.026	0.66	.105	2.67
.030	0.76	.112	2.84
.045	1.14	.120	3.05
.055	1.40	.124	3.15
.077	1.96	.140	3.56
.081	2.06	.145	3.68
.093	2.36		
.094	2.39		

Tools: See table II.

TABLE II. Tools.

BIN code	Standard fixture		Repair fixture	Heating tool	Installing and removal tool
	Basic fixture	Adapter			
399, 400, 401	AD - 1319 Raychem 1/	AT - 1319-19 Raychem 1/	AD - 1494 Raychem 1/	M83521/5-04	M81969/14-04

1/ Or equivalent.

Mating contact: MIL-C-39029/73.
Assembly instructions: Manufacturer's recommended assembly instructions shall be shipped with unit package. See figure 2.
(C) Solder type: SN63SRMAP2 in accordance with QQ-S-571.
(C) Cable accommodation. See tables I, IV, and V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:41Z
Check the source to verify that this is the current version before use.

Contact engagement and separation force. See table III.

TABLE III. Contact engagement and separation force.

Test pin diameter (inches)	Socket contact only					
	Initial			After conditioning		
	Minimum separation force	Maximum average engagement force	Maximum engagement force	Minimum separation force	Maximum average engagement force	Maximum engagement force
	oz	oz	oz	oz	oz	oz
.0295 (0.75) +.0000 (0.00) -.0001 (0.00)	N/A	N/A	16	N/A	N/A	20
.0280 (0.72) +.0001 (0.00) -.0000 (0.00)	0.5	N/A	N/A	0.4	N/A	N/A

Low signal level contact resistance: 20 milliohms, maximum. Tested with 24 AWG twisted pair.
Contact resistance. See table IV.

(C) TABLE IV. Contact resistance.

BIN code	Cable accommodated	Test current (amps) 1/	Maximum voltage drop (millivolts)					
			25°+3°,-0°C		25°+3°,-0°C 2/		125°+3°,-0°C	
			Inner	Outer	Inner	Outer	Inner	Outer
399	M17/095-RG180	1	120	13	144	16	204	22
400	M17/093-RG178 M17/094-RG179	1	120	13	144	16	204	22
	9350H1024-9 3/ (Raychem)	1	60	5	72	6	90	7.5
	24 AWG (twisted pair)	3	57	49	68	59	97	83
401	26 AWG (twisted pair)	2	63	51	76	61	107	87

1/ Ammeter accuracy of 2 percent.
2/ After conditioning.
3/ Or equivalent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:41Z
Check the source to verify that this is the current version before use.

Ⓒ Conductor retention (tensile strength test). See table V.

Ⓒ TABLE V. Conductor retention.

BIN code	Cables accommodated	Minimum tensile load 1/ (pounds)	
		Inner contact	Outer contact
399	M17/095-RG180	4.5	20.0
400	M17/093-RG178	5.2	6.5
	M17/094-RG179	5.2	16.0
	Raychem 9530H1024-9 2/	5.5	19.5
401	24 AWG twisted pair	15.0	15.0
	26 AWG twisted pair	10.5	10.5

1/ Breakage of cable outside of solder joint shall be considered passing.
2/ Or equivalent.

Vibration. Method 2005 of MIL-STD-1344, test condition VI, letter J, except the duration shall be 8 hours in the longitudinal direction and 8 hours in the perpendicular direction (16 hours total).

Shock (specified pulse): Method 2004 of MIL-STD-1344, test condition D.

Dielectric withstanding voltage: 900 V rms, minimum, at sea level.

Part number: See table VI.

TABLE VI. Part number.

BIN code	Part number	Supersedes
399	M39029/74-399	M39029/74-12A
400	M39029/74-400	M39029/74-12C
401	M39029/74-401	M39029/74-12B

Ⓒ QPL evaluating activity. 2750 Air Base Wing, Electronics Support Division (2750 ABW/ESP), Gentile AFS, OH 45444.

Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities.
Army - AR, MI
Navy - EC, MC
Air Force - 11, 99
DLA - ES

User activity.
Navy - OS

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5935-3509-2)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:41Z
Check the source to verify that this is the current version before use.